



PONTIFÍCIA UNIVERSIDADE CATÓLICA DE MINAS GERAIS

Teste de Desempenho - Teste de Software

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Resumo Executivo

Cenário de I/O — /checkout/simple:

A API suportou com folga 50 VUsers no Load Test (0% de erros, p95 = 299ms) e 300 VUsers no Spike Test (0% de erros, p95 = 299ms). A capacidade máxima utilizável é de aproximadamente 300 usuários simultâneos.

Cenário de CPU — /checkout/crypto:

A API colapsou já no primeiro stage do Stress Test. A capacidade máxima prática é de menos de 200 usuários simultâneos. Acima disso, o servidor entra em colapso total por timeout.

smoke.js:

```
PS C:\Users\caio\OneDrive\Documents\Faculdade\TesteSoftware\TrabalhoTesteDesempenho\teste-de-desempenho-caio-batella> k6 run .\tests\smoke.js

  execution: local
  script: \tests\smoke.js
  output: -

  scenarios: (100.00%) 1 scenario, 1 max VUs, 10m0s max duration (incl. graceful stop):
    * default: 1 looping VUs for 30s (gracefulStop: 30s)

  THRESHOLDS
    http_req_failed
      / 'rate<0.01' rate=0.00%

  TOTAL RESULTS
    checks_total.....: 67116 2272.00322/s
    checks_succeeded...: 100.00% 67116 out of 67116
    checks_failed.....: 0.00% 0 out of 67116
    ✓ status is 200
    ✓ status is 200

  HTTP
    http_req_duration.....: avg=738.73µs min=0µs med=688.4µs max=103.11ms p(90)=1.48ms p(95)=1.82ms
    { expected_response:true }.....: avg=738.73µs min=0µs med=688.4µs max=103.11ms p(90)=1.48ms p(95)=1.82ms
    http_req_failed.....: 0.00% 0 out of 33558
    http_reqs.....: 33558 1118.60361/s

  EXECUTION
    iteration_duration.....: avg=881.06µs min=0µs med=654.29µs max=240.8ms p(90)=1.67ms p(95)=2.23ms
    iterations.....: 33558 1118.60361/s
    vus.....: 1 min=1 max=1
    vus_max.....: 1 min=1 max=1

  NETWORK
    data_received.....: 9.7 MB 321 kb/s
    data_sent.....: 2.6 MB 85 kb/s

  running (0m30.0s), 0/1 VUs, 33558 complete and 0 interrupted iterations
  default ✓ [=====] 1 VUs 30s
  PS C:\Users\caio\OneDrive\Documents\Faculdade\TesteSoftware\TrabalhoTesteDesempenho\teste-de-desempenho-caio-batella>
```

load.js:

```
PS C:\Users\caio\OneDrive\Documents\Faculdade\TesteSoftware\TrabalhoTesteDesempenho\teste-de-desempenho-caio-batella> k6 run .\tests\load.js

  scenarios: (100.00%) 1 scenario, 50 max VUs, 4m0s max duration (incl. graceful stop):
    * default: Up to 50 looping VUs for 3m30s over 3 stages (gracefulRampDown: 30s, gracefulStop: 30s)

  THRESHOLDS
    http_req_duration
      / 'p(95)<500' p(95)=299.07ms

    http_req_failed
      / 'rate<0.01' rate=0.00%

  TOTAL RESULTS
    checks_total.....: 13718 65.21733/s
    checks_succeeded...: 100.00% 13718 out of 13718
    checks_failed.....: 0.00% 0 out of 13718
    ✓ status is 201
    ✓ status is APPROVED

  HTTP
    http_req_duration.....: avg=208.83ms min=99.85ms med=212.92ms max=328.92ms p(90)=204.82ms p(95)=299.07ms
    { expected_response:true }.....: avg=208.83ms min=99.85ms med=212.92ms max=328.92ms p(90)=204.82ms p(95)=299.07ms
    http_req_failed.....: 0.00% 0 out of 6859
    http_reqs.....: 6859 32.008967/s

  EXECUTION
    iteration_duration.....: avg=1.2s min=1.1s med=1.21s max=1.32s p(90)=1.28s p(95)=1.29s
    iterations.....: 6859 32.008967/s
    vus.....: 2 min=1 max=50
    vus_max.....: 50 min=50 max=50

  NETWORK
    data_received.....: 2.0 MB 9.6 kb/s
    data_sent.....: 1.2 MB 5.8 kb/s

  running (3m30.3s), 0/50 VUs, 6859 complete and 0 interrupted iterations
  default ✓ [=====] 0/50 VUs 3m30s
  PS C:\Users\caio\OneDrive\Documents\Faculdade\TesteSoftware\TrabalhoTesteDesempenho\teste-de-desempenho-caio-batella>
```

stress.js:

```
WARN[0361] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": request timeout"
WARN[0361] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": request timeout"
WARN[0361] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": request timeout"
WARN[0361] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": request timeout"
WARN[0361] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": request timeout"
WARN[0361] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": request timeout"
WARN[0361] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": request timeout"

THRESHOLDS
http_req_failed
  / "rate0.10" rate=99.02%

TOTAL RESULTS
checks_total.....: 307337 787.990635/s
checks_succeeded.....: 0.97% 2994 out of 307337
checks_failed.....: 99.02% 304343 out of 307337

X status is 201
  0% - / 2994 / X 304343

HTTP
http_req_duration.....: avg=242.67ms min=0ms      med=0ms      max=1ms      p(90)=0ms      p(95)=0ms
{ expected_response:true }.....: avg=13.5s      min=65.28ms med=6.00s      max=59.99s      p(90)=35.41s      p(95)=52.1s
http_req_failed.....: 99.02% 304343 out of 307337
http_reqs.....: 307337 787.990635/s

EXECUTION
iteration_duration.....: avg=480.5ms min=0ms      med=215.95ms max=1ms      p(90)=502.90ms p(95)=715.57ms
iterations.....: 307337 787.990635/s
vus.....: 30 min=2      max=999
vus_max.....: 1000 min=1000      max=1000

NETWORK
data_received.....: 940 KB 2.4 kb/s
data_sent.....: 787 KB 2.0 kb/s

running (6m30.0s), 0000/1000 VUs, 307337 complete and 29 interrupted iterations
default [=====] 0025/1000 VUs 6m0s
ps[0] (0360) thresholds on metrics 'http_req_failed' have been crossed
PS C:\Users\caio\OneDrive\Documents\Faculdade\TestesSoftware\TrabalhoTestesDesempenho\teste-de-desempenho-caio-batella > []
```

spike.js:

```
PS C:\Users\caio\OneDrive\Documents\Faculdade\TestesSoftware\TrabalhoTestesDesempenho\teste-de-desempenho-caio-batella > k6 run .\tests\spike.js

execution: local
script: .\tests\spike.js
output: -

scenarios: (100.00%) 1 scenario, 300 max VUs, 2m0s max duration (incl. graceful stop):
  * default: up to 300 looping VUs for 1m50s over 4 stages (gracefulRampDown: 30s, gracefulStop: 30s)

THRESHOLDS
http_req_failed
  / "rate0.05" rate=0.00%

TOTAL RESULTS
checks_total.....: 17700 159.122019/s
checks_succeeded.....: 100.00% 17700 out of 17700
checks_failed.....: 0.00% 0 out of 17700

✓ status is 201

HTTP
http_req_duration.....: avg=208.32ms min=95.40ms med=206.43ms max=775.02ms p(90)=288.91ms p(95)=299.02ms
{ expected_response:true }.....: avg=208.32ms min=95.40ms med=206.43ms max=775.02ms p(90)=288.91ms p(95)=299.02ms
http_req_failed.....: 0.00% 0 out of 17700
http_reqs.....: 17700 159.122019/s

EXECUTION
iteration_duration.....: avg=1.21s min=1.09s med=1.2s      max=1.86s      p(90)=1.28s      p(95)=1.3s
iterations.....: 17700 159.122019/s
vus.....: 1 min=1      max=300
vus_max.....: 300 min=300      max=300

NETWORK
data_received.....: 5.2 MB 47 kb/s
data_sent.....: 3.2 MB 29 kb/s

running (1m51.1s), 000/300 VUs, 17700 complete and 0 interrupted iterations
default [=====] 000/300 VUs 1m50s
PS C:\Users\caio\OneDrive\Documents\Faculdade\TestesSoftware\TrabalhoTestesDesempenho\teste-de-desempenho-caio-batella > []
```

Análise de Estresse

O ponto de ruptura foi em torno de 200 VUsers (início do primeiro stage do stress.js).

O motivo é que o endpoint /checkout/crypto usa bcrypt.hashSync(), que é síncrono e bloqueia completamente o Event Loop do Node.js. Como o Node.js é single-threaded, cada requisição ocupa a CPU inteiramente até terminar — fazendo com que as demais fiquem na fila até dar timeout.

Ao atingir ~200 VUsers simultâneos, o servidor já não conseguia processar as requisições a tempo, resultando em 99,02% de falhas e timeouts generalizados ao escalar para 1.000 VUsers.